AI Primer: Getting Started with Large Language Models and Building AI Agents

Artificial Intelligence (AI) systems like ChatGPT and Google Gemini are powered by Large Language Models (LLMs), which are mathematical prediction engines trained on vast amounts of data from the internet and elsewhere, enabling the 'learning'** of complex grammar, context and factual patterns.

The Task Horizon for AI is accelerating rapidly. Within a few years, interacting with software and machines using everyday language is set to become the default experience. Instead of navigating complex user interfaces or relying heavily on manual mouse and keyboard controls, users will simply talk to applications.

To effectively command this new era, managers, architects and builders must learn how to properly steer these models. By providing specific instructions and tweaking internal settings, AI behaviour can be focused from wildly creative to strictly factual. Turning these predictive engines into reliable tools relies on three core skills:

1. Prompt and Context Engineering: Steering models with precise instructions and constraints while curating the exact background data required to ground the AI in reality. Tutorials: Prompt Engineering Full Course and Context Engineering for LLMs
2. Retrieval-Augmented Generation (RAG): Connecting the AI to external information like network folders. This extracts factual answers from inaccessible and unstructured data. Tutorial: RAG Explained in 20 Minutes
3. AI Agents: Equipping models with external tools to complete complex workflows. Tutorial: AI Agents for Beginners

AI Large Language Models (LLMs) are highly adept at processing, analysing, summarising, translating and generating information, often in 'real time' (e.g. 'Chatbots'). In addition, otherwise unstructured information such as documents stored in network folders, email archives, sprawling information such as systems logs and historic information repositories are significantly more accessible and actionable.

However, it is important to avoid the trap of over-reliance on current AI capabilities. While these tools are excellent for specific tasks, deploying models indiscriminately or without human oversight often leads to poor outcomes. Lazy integrations and misjudged use cases will fail to deliver promised cost savings or operational efficiencies. Because AI operates purely as a statistical prediction engine lacking genuine comprehension or human intuition, AI must augment rather than replace human judgement.

How Large Language Models Actually Work

**Operating as advanced statistical prediction and pattern recognition engines, the models calculate the most likely next word in a sequence, functioning like a highly sophisticated autocomplete. For a concise visual overview, watch Large Language Models Explained Briefly.

By mapping the mathematical patterns of human interaction, these models closely replicate natural conversation, creating a powerful illusion of intelligent reasoning. Despite this convincing illusion, Artificial Intelligence has no genuine comprehension of the topics discussed, entirely lacking human intuition or self-awareness. Furthermore, LLMs do not actively store facts, files or images like traditional databases.

Instead, training data is encoded purely as billions (or trillions) of numerical parameters, known as weights and biases. Consequently, models generate responses based on probability rather than true knowledge. Through sophisticated mathematical techniques, massive enterprise models can be compressed to as small as 2GB for local use and perform well if used correctly for specific tasks.

The AI Toolkit: Apps and Platforms

A wide range of free high quality software and training resources are available. The four software groups below scale from software than can be downloaded and installed on Laptops or Desktop PCs, to 'free for personal use' Cloud Hosted Enterprise Tools for building AI Agents and Workflows.

Alternatively, subscription services such as ChatLLM and OpenRouter provide access to the latest premium models from Meta, Google and other leading AI companies. These services for circa £10 per month, usually include Tools and API access for application development. Microsoft 365 Copilot integrates with Microsoft Azure and Office applications from around £20 per month.

Group A: On-Premises Tools (Local Installation)

These applications are free to download. Reasonably modern computers with 8GB of RAM will run smaller models locally, though a dedicated graphics card (6GB+ VRAM) is recommended.

- LM Studio: Local environment for prompt engineering. Tutorial: LM Studio Setup.
- AnythingLLM: Desktop app for secure, local document processing. Tutorial: AnythingLLM RAG.

Group B: Hosted Platforms for Prompt and Context Engineering

These platforms facilitate both Prompt Engineering (crafting instructions) and Context Engineering (curating specific background data to ground the AI).

- Google AI Studio: Free web testing environment showing how visual sliders alter underlying code while allowing users to upload specific context files. Tutorial: Google AI Studio Beginners.
- prompt.new: Distraction-free web application specifically built for refining instructions and testing context structures. Tutorial: Portkey Prompt Studio.

Group C: API and Python Development

- Google Colab: Free cloud notebook to run Python code without local setup. Tutorial: Google Colab Beginners.
- OpenAI and Anthropic SDKs: Write raw commands using Python. Tutorial: OpenAI API Tutorial.
- LangChain (Python): Framework for complex document retrieval and agents. Tutorial: LangChain Crash Course.

Group D: AI Agents and Enterprise Workflows

- Flowise, Coze and Dify.ai: Drag-and-drop interfaces for building custom AI Agents and managing document retrieval. Tutorials: Flowise Setup, Coze Starter and Dify Quickstart.
- n8n Cloud, Gumloop and Make.com: Visual flowcharts connecting models to business tools for zero-code automations. Tutorials: n8n Assistant, Gumloop Automations and Make.com Guide.
- Claude and Claude Code: Autonomous command-line tool bridging the gap to enterprise development. Tutorial: Claude Code Overview.

AI Terminology and Glossary

Core Concepts and Interaction

- **System Prompt**: The foundational instruction set defining the AI's core persona, behaviour, and formatting rules before greeting the user.
- **User Prompt**: The actual query, instruction, or task entered into the chat interface.
- **Token**: The fundamental unit of data processed by an AI, roughly equivalent to a single word or a fragment of a word.
- **Context Window**: The AI's short-term memory limit. This metric dictates the maximum number of tokens the model can remember during a single continuous conversation.

LLM Control Parameters

- **Temperature**: Controls mathematical randomness. A 0.1 value makes output highly factual. A 0.7 value provides conversational balance. A 1.5 value makes output wildly creative, risking incoherence.
- **Top-P**: Limits vocabulary by cutting off least likely choices via percentage. A 0.1 value forces robotic text, while 0.9 allows richer vocabulary.
- **Top-K**: Limits vocabulary to a strict numerical limit of top words. A value of 1 guarantees repetitive loops, 40 provides safe balance and 100+ increases logical errors.
- **Frequency Penalty**: Penalises reusing exact words. A 0 value allows natural repetition, while 2 aggressively forces synonyms, often breaking grammatical rules.
- **Presence Penalty**: Penalises staying on the exact same topic. A 0 value keeps focus, while 2 forces rapid, often chaotic topic switching.

Advanced Architecture

- **Retrieval-Augmented Fine Tuning (RAFT)**: A training technique improving upon standard RAG. It actively teaches AI how to identify and ignore irrelevant distraction documents when searching for answers, ensuring precision.
- **Speculative Decoding**: An inference speed boost. A smaller, faster AI model guesses the next few words in a sequence and the primary, larger AI model instantly verifies the output.